بسم الله الرحمن الرحيم

ENDOCRINAL DISORDERS

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Hypothyroidism

- **Thyroid hormone is essential for normal somatic growth and neurological development in infants and children. It is also necessary for normal skeletal growth in growing children. Deficiency of thyroid hormone in the first 2 years of life may result in physical and psychomotor retardation.**

- So the clinical manifestations differ in infants and older children or adolescents.
- In children and adults, thyroid hormone plays a major role in metabolism and heat production.
- Hypothyroidism may occur at birth (congenital hypothyroidism) or at any time during childhood or adolescence (juvenile hypothyroidism).

Congenital hypothyroidism

Etiology:

1- ***Developmental thyroid defects (thyroid agenesis or dysgenesis):*** **these are the primary causes of congenital hypothyroidism.**

2- ***Defective biosynthesis of thyroid hormone:*** **it frequently results in goiter and may cause hypothyroidism.**

Clinical features

Neonatal period

- **Birth weight and length are normal, but head size may be slightly increased because of myxedema of the brain.**
- **Prolongation of physiologic icterus due to inadequate bilirubin conjugation may be the earliest sign.**

- **Feeding difficulties e.g. somnolence and chocking spells during nursing.**
- **Respiratory difficulties due to large tongue including apneic episodes, noisy respirations, and nasal obstruction.**

Infants and children

- **Physical and mental retardation which increases with age.**
- **Affected infants cry little, sleep much, have poor appetite, and are generally sluggish.**
- **Hoarseness of voice.**
- **There may be constipation that does not usually respond to treatment. The abdomen is large, and an umbilical hernia is usually present.**

- **The temperature is subnormal and the skin is cold and mottled.**
- **The pulse is slow, heart murmurs and cardiomegaly are common.**
- **The child's growth is stunted, and the extremities are short.**
- **Broad hands and short fingers.**

- **Laboratory data**

• Serum T_3 may be normal and is not helpful in diagnosis.

• Serum T_4 level is low. • High TSH, if the defect is primarily in the thyroid.

- ***Screening newborns for congenital hypothyroidism:*** **The initial test performed for T4 in whole blood at day from 3-7. Infants with low result are considered suspect.**

- **Treatment**

  Oral L-thyroxin is the treatment of choice for life. The dose is about 10-15 µg/kg/24hr in neonates, 4 µg/kg/24hr in children, and in adults only 2 µg/kg/ 24hr. Thyroxin tablets should not be mixed with soy protein formula or iron, because these can inhibit its absorption. Levels of both T_4 and TSH should be monitored and maintained in the normal range. If toxicity occurs, the dose is reduced.

Diabetes mellitus

Diabetes mellitus (DM) is a chronic metabolic syndrome characterized by hyperglycemia, and is caused by a deficiency of insulin secretion and/or insulin action resulting in abnormal metabolism of carbohydrate, protein and fat.

Types of diabetes mellitus

I- Type I, juvenile-onset, insulin dependent diabetes mellitus (IDDM):

- **Individuals affected by this type confront serious burdens that include an absolute daily requirement of exogenous insulin, with a need to monitor their own metabolic control and to pay a constant attention to dietary intake.**
- **There is severe insulinopenia with dependence on exogenous insulin to prevent ketosis and to preserve life.**

- **The onset occurs predominantly in childhood. Peaks of presentation occur in two age groups: at 5-7 years and at the time of puberty.**

- ***II- Type II, adult-type, non insulin dependent diabetes mellitus (NIDDM):***

- **This type is not insulin dependent.**
- **It infrequently causes ketosis except in severe infection and stress.**
- **Many persons of this category are obese and inactive.**

Juvenile onset diabetes

- **It is due to decreased secretion of insulin.**
- **An autoimmune destruction of pancreatic islets in predisposed individuals is suggested to be an etiologic factor.**
- **Triggering factors include viral infections (e.g. mumps, rubella). The virus may act by direct destruction of beta cells or by triggering wide spread immune response to several endocrine tissues.**

Pathophysiology

- The progressive destruction of beta cells leads to progressive deficiency of insulin which is a major anabolic hormone.
- Excessive hyperglycemia leads to glucosuria when the renal threshold of 180mg/dL is exceeded.

- **There is accelerated lipolysis and impaired lipid synthesis with increased plasma concentration of total lipids, cholesterol, triglycerides and free fatty acids.**
- **Increased formation of ketone bodies by dissociation of the free fatty acids in the blood can cause ketonuria and ketonemia. This results in metabolic acidosis with rapid deep breathing and leads to impaired consciousness and coma.**

Clinical manifestations

- Polyuria and polydypsia with polyphagia and weight loss.
- Insidious onset of lethargy and weakness.
- Pyogenic skin infections and monilial vaginitis in teenage females.

- ***Ketoacidosis*** **can be the initial presentation in approximately 25% of diabetic children. It is manifested by:**

a- Vomiting, polyuria, dehydration, and coma.

b- Metabolic acidosis with deep rapid respiration.

c- Acetone odor of the breath.

d- Acute abdomen may be present and mimic surgical conditions.

- **Laboratory diagnosis of diabetes mellitus:**

- **Glucosuria and ketonuria plus a random plasma glucose ≥200 mg/dL or fasting plasma glucose ≥ 126 mg/dL or two-hours plasma glucose during oral glucose tolerance test ≥ 200 mg/dL.**

- **Laboratory features of diabetic ketoacidosis include:**

a- Blood glucose level more than 300 mg/dL.

b- pH level less than 7.3 (normal = 7.35-7.45).

c- Serum bicarbonate (HCO_3) less than 15 mmol/L (normal level = 24 mmol/L).

- **Complications**

Retinopathy, nephropathy, neuropathy, dwarfism, enlarged liver, osteopenia and psychologic disturbances.

Treatment of IDDM

- **The management of IDDM may be divided into three phases depending on the initial presentation:**

A- Ketoacidosis.

B- Post acidotic phase or transition period for establishment of metabolic control.

C- Continuing phase of control and guidance of the diabetic child and his or her family.

- *A- Ketoacidosis*

The immediate aims of therapy are expansion of intravascular volume, correction of deficits in fluids, electrolytes and acid-base status, and initiation of insulin therapy.

- **Expansion of the reduced intravascular volume and correction of depleted fluid and electrolyte stores are most important in treatment of diabetic ketoacidosis.**
- **The degree of dehydration is commonly about 10%.**
- **The initial hydration fluid should be isotonic saline (0.9%).**

- **The rate of fluid replacement is adjusted to provide only 50%-60% of calculated deficit within the initial 12 hours and the remaining 40%-50% is administered during the next 24 hours.**
- **Administration of glucose 5% solution in 0.2% NaCl is initiated when blood glucose concentration approaches 300mg/dL.**
- **Correction of acidosis:Sodium bicarbonate therapy is recommended only when pH is 7.2 or less**

- **Insulin therapy**

  Fast-acting insulin (crystalline) is started with initiation of fluid therapy.

- **When acidosis has been corrected and if blood glucose is approaching 300mg/dL, insulin is given subcutaneously in a dose of 0.2-0.4 U/kg every 6 to 8 hours while maintaining the glucose infusion until the child can fully tolerate food.**

- **The blood glucose level should be monitored before and 2 hours after each meal and the insulin dose adjusted to maintain the blood glucose concentration in the range of 80-180 mg/dL.**
- **The total dose of regular insulin used in this representative day serves as a guide for subsequent insulin treatment with a combination of intermediate and short acting insulin. The patient at the end of this stage should be well hydrated and fully conscious.**

- **During insulin therapy, the patient should be observed for hypoglycemia that is manifested by irritability, convulsions, sweating, dilated pupils, epigastric pain and coma and should be treated by IV glucose 25%.**

- ***B- Post acidotic or transition period***

- **Diabetic ketoacidosis is usually corrected within 36to 48 hours by the foregoing therapeutic regimen. At this time, food and fluids are usually tolerated orally, and insulin can be given by subcutaneous injection.**

- **The child who presents with the classic symptoms with documented hyperglycemia, in the absence of clinical dehydration and ketoacidosis, can be considered as requiring treatment at this transition stage. For such children, subcutaneous injections of fast acting insulin are begun at doses of 0.1-0.25 U/kg every 6-8hr before meals with simultaneous monitoring of blood glucose concentration**

and adjustment of the insulin dose for 1-2 days and these doses are used as a guide to subsequent use of intermediate and short acting insulin. For newly diagnosed patients, this phase commonly lasts 5-10 days. Ongoing education and adjustment of insulin dosage are continued after discharge from the hospital through patients' visits.

- ***C- Continuing control and guidance***

1-Insulin regimens

- **One of the following regimens may be used:*Single daily dose:* a fast acting e.g. regular insulin (⅓ of the dose) is combined with intermediate acting insulin, e.g. NPH, lente (⅔ of the dose). The injection is given 30 min. before breakfast. The two types of insulin should be drawn into the syringe in the same sequence (regular first).**

- **It is indicated in:**

1- Children aged more than 5years and at puberty.

2- No fasting hyperglycemia.

3- No nocturnal enuresis.

- ***Double daily doses:*** are now routinely recommended dividing the total daily dose into two injections, ⅔ of the calculated dose is given before breakfast and the other ⅓ is given 30minutes before the evening meal and each injection consists of ⅓ of the dose as a regular insulin and ⅔ of the dose as an intermediate acting insulin. It is indicated in:
- 1- Age below 5 years.
- 2- Fasting hyperglycemia.
- 3- Nocturnal enuresis.

- ***Three or more daily injections of insulin:*** based on frequent blood glucose monitoring, can be considered in special circumstances such as highly motivated older children or adolescents or with capable parents. This kind of intensive insulin therapy improves glycemic control toward normal, diminishes the risk of microvascular complications, but is often associated with more severe and frequent hypoglycemic reactions.